

Day 1

Welcome – Col Kraus (15 mins)

Welcome/Logistics – Dr. Roach/Dr. Bin-Salamon (5 mins)

- I. University of Maryland
 - a. Overview – Losert (30 mins)
 - b. Introduction of Analysis – Losert (30 mins)
 - c. Electrotactic signaling pathways - Devreotes (30 mins)

BREAK (10 mins)

- d. Novel electric field generators and probes – Quan (30 mins)
- e. Controlling excitable networks and signaling pathways with nanotextures – Fourkas (30 mins)
- f. Electrotaxis in epithelial cells and tissues – Zhao (30 mins)

BREAK (1.5 hours)

- g. Electrical excitable networks – Kanold (30 mins)
- h. Review of MURI Aims – MURI Team (30 mins)

Day 2

Welcome/Logistics – Dr. Roach/Dr. Bin-Salamon (5 mins)

- II. Old Dominion University
 - a. Overview – Pakhomov (15 mins)
 - b. Engineering Core Report: Novel nanosecond pulse generators and delivery systems aimed at studying cancellation effects – Xiao (30 mins)
 - c. Bipolar pulse cancellation: The role of pulse shape and duration; Peripheral nerve excitation by nanosecond-duration stimuli - Pakhomov/Casciola (30 mins)
 - d. Spectral analysis of transmembrane potential induced by unipolar and bipolar pulsed electric fields; Sub-ten-nanosecond bipolar pulse cancellation – Vernier (30 mins)
 - e. Overview of mechanistic nanopore modeling for reversal; MD results can improve cell models; Cell system modeling with transient and long-lived nanopores – Stern/Weaver (30 mins)
 - f. Non-invasive assessment of electrical and mechanical properties on a microscale – Yakolev (30 mins)
 - g. Anoctamin 6 is involved in plasma membrane phospholipid perturbation following nsPEF insult; Assisted membrane discharge: A potential mechanism of bipolar pulse cancellation – Pakhomova/Muratori/Bruno (30 mins)
 - h. Response of excitable adrenal chromaffin cells to monopolar and bipolar pulses – Craviso/LeBlanc (30 mins)

i. General discussion and next year targets (30 mins)

Biophysics Presentation – Dr. Roach/ Dr. Bin-Salamon (30 mins)

Open discussion – Dr. Roach (30 mins)

Government/Reviewer closed discussion